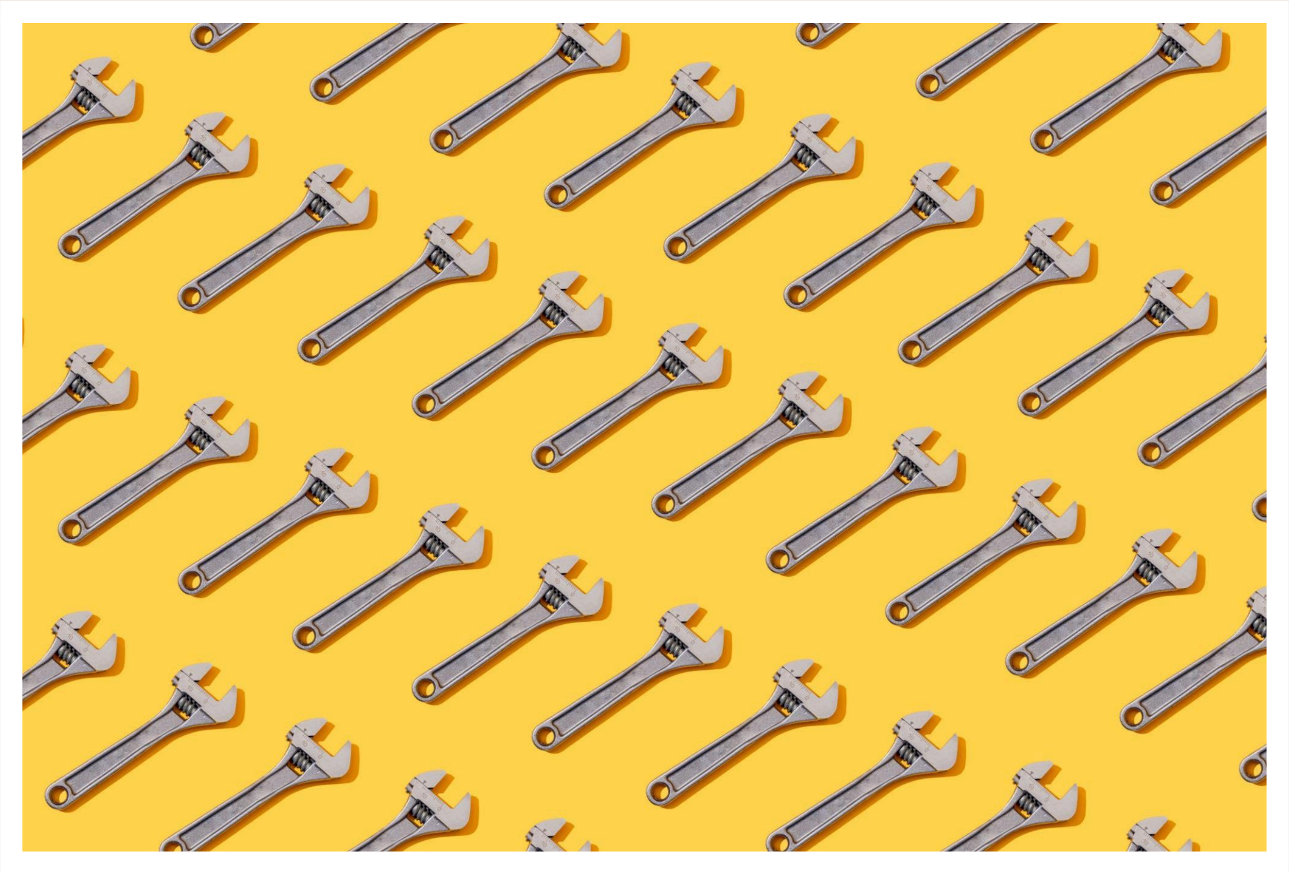
# Aipermind User's manual

app.aipermind.com



01 - Introduction
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# Technical guide

For testing on app.aipermind.com

01

## Introduction

### What is Aipermind?

Aipermind is a multi-agent AI platform for testing and validating ideas, product|service|business concepts, through interaction with Customer Digital Twins.

# Benefits

→ 24/7 availability for interactions and feedback

→ Ability to easily scale searches

→ Reduction of costs and research times

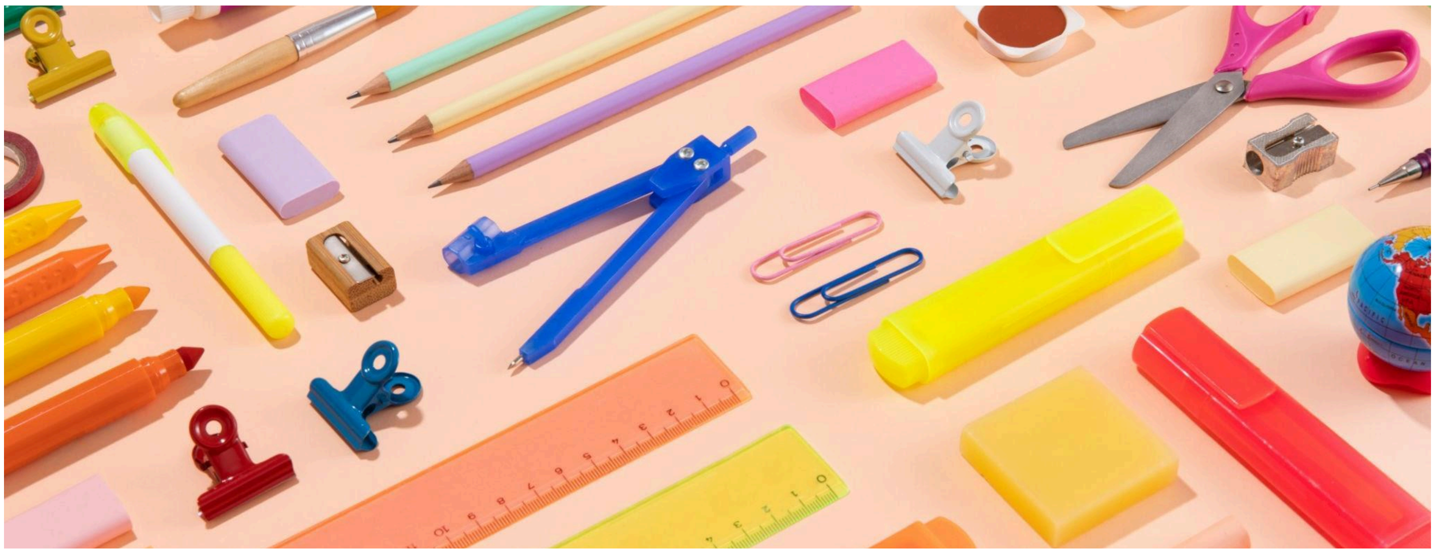
→ Speed in getting feedback

→ Daily practice of *User Centricity*

→ Democratization of the *Voice of Customer* : on everyone's desk, accessible and professionally distilled

→ From *guessing* to *feedback* : it is possible to create the Digital Twin of each stakeholder, external or internal to the decision-making process, and refine each proposal before delivery

# SYSTEM REQUIREMENTS



The platform can be accessed from the web address app.aipermind.com with username and password

| | |
|---|---|
| → PC/Mac Desktop or Tablet | → Good internet connection |

02

# Digital Twins

### MEANING

## What are they?

A Digital Twin is a virtual representation of a person, object, or system that exists in the real world.

We can think of it as a "digital mirror" that reflects the characteristics, behaviors and interactions of the original.

In the field of virtual representations of people, we talk about **Human Digital Twin** or **Customer Digital Twin** . In this document we will only talk about Human/Customer Digital Twin, and therefore for simplicity we will only say Digital Twin or **Twin** .

→ **Real Digital Twin**

- It is a digital replica that mirrors a real-life person or system.
- Maintains a direct and continuous connection with its physical "twin"

→ **Synthetic Digital Twin**

- It is an artificially created virtual representation
- It has no direct counterpart in the real world
- It is generated by combining data and behavioral patterns from different sources

- It is updatable, based on data and interactions of the real counterparty
- It is used to analyze and predict the behavior of the original

- It is used to simulate scenarios and test interactions in a controlled environment.

## CHARACTERISTICS AND GENERAL PRINCIPLES

### TRANSPARENCY AND ANONYMY

➔ The Digital Twin maintains confidentiality of sensitive data, which is in no way disclosed during interactions (nor stored in any system)

➔ During modelling, the profile is anonymised and stripped of any biometric trace (e.g. tone of voice) that can be traced back to the real 'matrix'

### ACCURACY, CONSISTENCY

➔ The answers that DTs provide must be detailed and precise.

➔ The information that DTs provide is based on - and supported by - hard data.

➔ The behavior is always consistent with the original profile

### QUALITY OF INTERACTION

➔ The tone of communication is positive and constructive, but in no way condescending.

➔ The answers are relevant to the context and are not influenced by the interlocutor's expectations

- The twin must be able to handle both agreement and disagreement appropriately.

- In order to always maintain a high level of quality in the interaction, the replies only keep memory of the ongoing conversation.

**CONTINUOUS MONITORING**

- The twin's performance is evaluated regularly, according to the principles of *observability* .[1]

- Metadata on interactions are periodically analyzed to improve the service and ensure high quality standards, without in any way entering into the merits of the content

These criteria ensure that digital twins, both real and synthetic, maintain high standards of quality and reliability in their interactions, ensuring an optimal and safe user experience.

[1] https://www.datadoghq.com/knowledge-center/llm-observability/

# Reliability of Digital Twins

## Real Digital Twins are associated with a reliability index , which strictly depends on the way in which the source data is collected, and which is observable through the transcript of the interview underlying the modelling [2].

The reliability score is a percentage measure that indicates how accurate and reliable a Digital Twin is. This score is calculated by considering several key factors:

**PROFILE COMPLETENESS**

- ➔ Facts and experiences: how many significant experiences and events are documented
- ➔ Behavioral Traits: How well defined are characteristic behaviors
- ➔ Demographics: Evaluate the completeness of demographic information that helps contextualize the Twin in its environment

**QUALITY OF TRANSCRIPTION AND INTERVIEW**

- ➔ Clarity of Language: Evaluates how clear and understandable the information is. Considers the precision of communication
- ➔ Quality of Questions: Measure how open and productive the questions were during the interview. Check for the absence of questions that were too convergent, or tendentious, or hypothetical (which worsen the quality of the data collected)
- → Depth of Responses: Analyze how detailed and complete the responses are. Consider the level of depth of the information
- → Balance the Conversation: evaluates the balance between questions and answers, in favor of the answers. It is important that the interviewee dominates the conversation, and that the interviewer limits himself to asking short questions.

The reliability score is not static, but can change over time, in the case of subsequent data enrichment [3]:

- It updates when new information is added
- Improve with profile enrichment
- Reflects the "freshness" and relevance of the data

This scoring system:

- Help users understand how much they can “trust” the information that emerges from interacting with Digital Twins

[2] https://arxiv.org/abs/2411.10109 - Researchers at Stanford University and Google DeepMind have conducted research that discusses in detail the principles behind the modeling and reliability of Human Digital Twins.

[3]Single Digital Twin enrichment is a feature in beta-testing.

- Provides the user with clear feedback on how to maximize the quality of Twins in their library
- It introduces a level of observability and accountability on the part of the user.

# USE

Digital Twins are innovative tools that allow the creation of interactive virtual representations of real or synthetic people, with multiple practical uses:

→ **Natural and personalized interaction through Chat**

- Digital Twins can have natural conversations with the user through a chat
- Each Twin has its own personality and unique set of experiences.
- The answers are contextualized and consistent with the Twin's profile
- Interaction is limited to a maximum number of exchanges to preserve a maximum degree of information quality, and to keep the interaction focused on one learning objective at a time.

→ **Market Test**

- Digital Twins can participate in structured interviews, in batches (so, for example, a test can include 10 individual interviews with 10 different twins, with the same objective)
- Tests are used for:
  - Evaluate new products or solutions
  - Identify problems and challenges
  - Exploring market opportunities
  - Collect detailed feedback
- The interviews follow a professional format because they are mediated by a properly trained and monitored virtual interviewing agent.

## UTILITY'

### Practical uses of Digital Twins

- Preliminary market research
- Concept and Idea Testing
- Validation of business hypotheses
- Understanding user needs

### Advantages

- 24/7 availability for interactions and feedback
- Ability to easily scale searches
- Reduction of costs and research times
- Speed in getting feedback
- Candidness: They are not subject to the typical human biases, and give 'sincere' feedback.

## FAQ

→ **What is the difference between the Synthetic Twin and the Customer Persona?**

[TL%DR] The Synthetic Twin is dynamic and autonomous. Its feedback is not mediated by the designer's interpretation. The Customer Persona is static and always 'expresses' itself through the designer's interpretation.

Extended answer

The Customer Persona is a static archetypal representation of a set of real people who share some key characteristics (for example, the Customer Persona can represent a target market). Thanks to these characteristics, the Customer Persona is used as an 'alias' of the real customer/referent. Being a static alias, its use is mediated by the designer. It is the designer who makes the Customer Persona 'speak', and who establishes whether what he has designed is consistent or not with the profile.

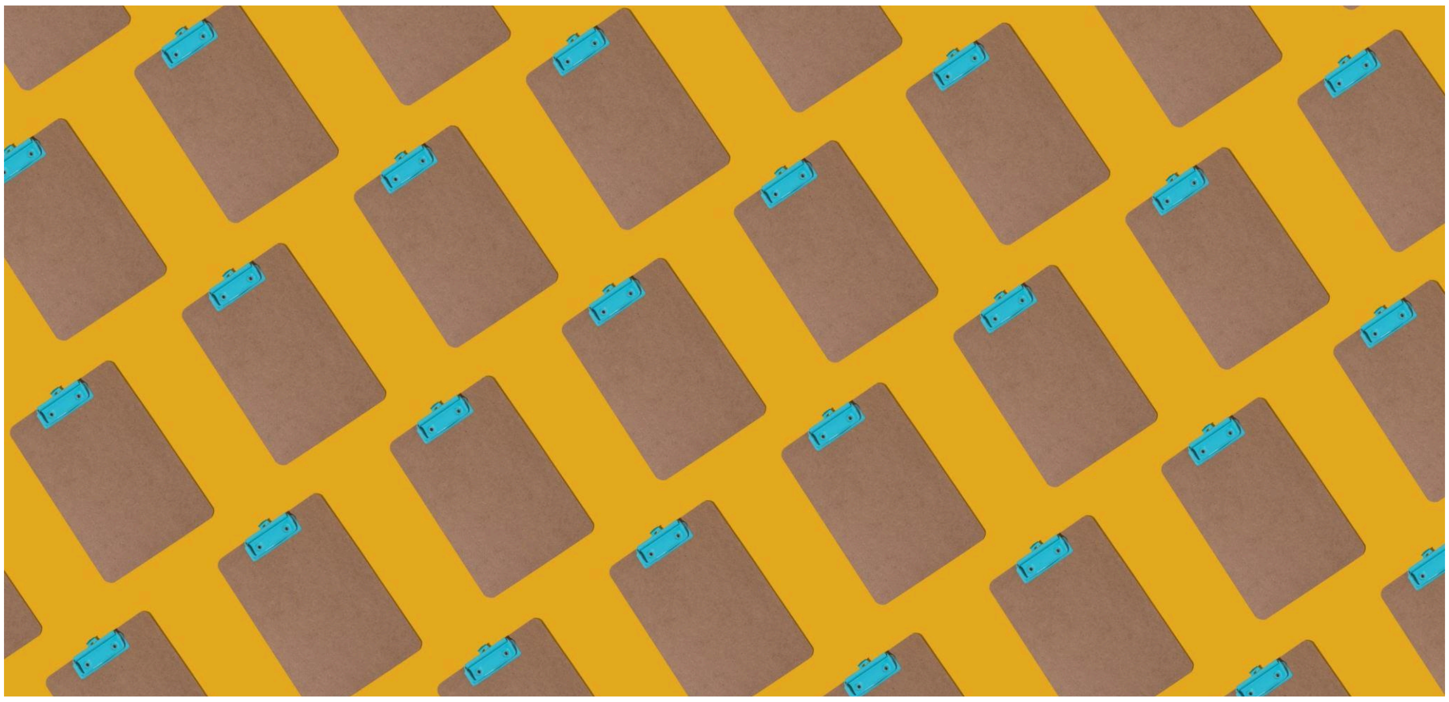
The Synthetic Digital Twin, on the other hand, is responsible for providing feedback that is consistent with its behavioral model, without demonstrating gratuitous condescension towards the designer. In this way, the designer obtains transparent, frank and objectified feedback with respect to his expectations, without having to make the effort (often compromised) to 'put himself in the shoes of' a person who could/should question his choices.

→ **What is the difference between Chat GPT (or other generic Chat-based LLMs)?**

[TL%DR] Asking Chat GPT/Gemini/Claude to 'act like' a certain person (providing any description) is feasible, but not effective and not useful in a systematic testing process. The interaction produces compliant, non-reproducible and therefore unreliable results, because it is not preceded by an effective 'modeling' process.

# 03

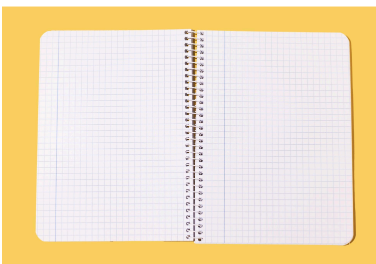
# Tests



**MEANING**

## What is a test?

A Test consists of a series of interviews, with a single learning objective. The interviewees are Digital Twins, and the interviewer is a digital agent trained to professionally manage each of the following types of tests.

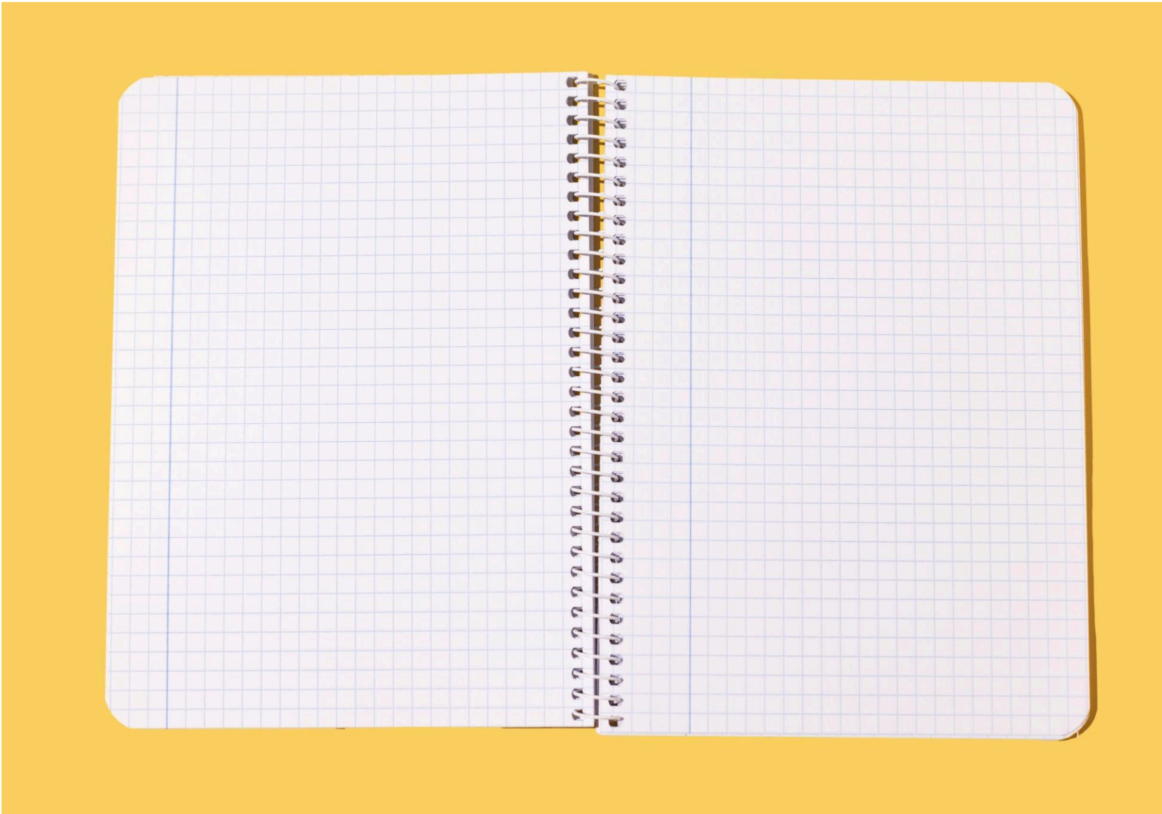


Market validation

Solution Validation

Custom Interview

# MARKET VALIDATION TEST



### FOCUS OBJECTIVE

- Understanding problem-market fit
- Combine problem exploration with solution validation

### STRUCTURE USE

- Semi-structured investigative interview by an experienced interviewer
- Ideal for early stage projects that need to explore the market space

## SOLUTION VALIDATION TEST

### FOCUS OBJECTIVE

| | |
|---|---|
| → Validate a specific solution | → Clear presentation of the solution concept<br>→ Collect detailed feedback on every aspect<br>→ Identifying barriers to adoption<br>→ Comparison with existing alternatives<br>→ Pricing and Value Perception Exploration |

### STRUCTURE USE

| | |
|---|---|
| → Semi-structured investigative interview by an experienced interviewer | → Clear presentation of the solution concept<br>→ Collect detailed feedback on every aspect<br>→ Identifying barriers to adoption<br>→ Comparison with existing alternatives<br>→ Pricing and Value Perception Exploration |

## CUSTOM TEST

### FOCUS OBJECTIVE

- Custom interviews with user-defined script
- Flexible according to specific needs

### STRUCTURE USE

- It follows freely a personalized script provided by the interviewer. The questions provided by the interviewer are declined on the basis of the characteristics of the interviewee and the progress of the conversation.
- From expert market researchers, who know exactly what to ask and how
- When you have specific questions to ask the interviewees that are not covered by the previous tests
- Provide 5-8 questions (there are 10 conversation turns)

# Step by step instructions

## How to set up, launch and use a test

**1. SETUP**

- ➔ Test name definition (you can change it later)

- ➔ Select the type of test (Market, Solution or Custom Validation)

- ➔ Configuring specific parameters for the chosen test type

**2. PARTICIPANTS**

- ➔ Selection of participants from the list of Digital Twins

**3. REVIEW**

- ➔ Review of the launch plan

**4. LAUNCH**

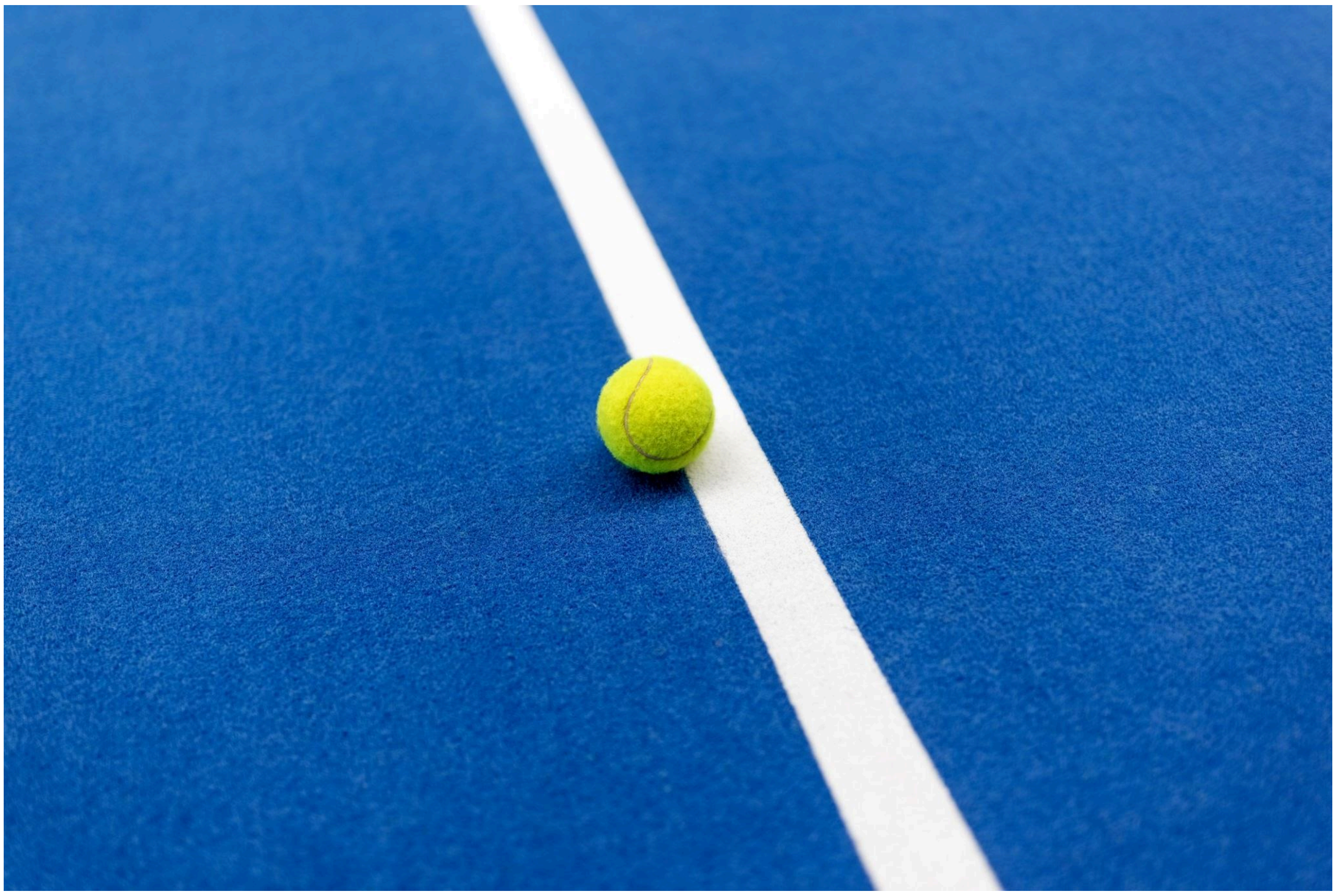
- ➔ The interview is conducted immediately and asynchronously: the duration is variable (in minutes), and the results are delivered both through the platform and through an email notification to the user.
- ➔ In the meantime, the user can devote himself to other activities on the platform or elsewhere (he can also close the PC, it does not interfere with the completion of the process)

**5. RESULTS**

- ➔ The report of each test consists of:
    - Summary report, the structure of which depends on the type of test and its objective
    - Report of each single interview
    - Transcript of each individual interview

- The report can be viewed on the Aipermind platform once the test is completed, and can be downloaded/shared in Word format.

# Clustering and Resegmentation



Clustering refers to the identification, on the basis of the transcripts of each interview of a Test, of internally homogeneous and externally heterogeneous subsets of responses.

Clustering is in fact a bottom-up re-segmentation, because it allows to identify sub-segments of Digital Twins (and therefore their counterparts) with respect to the topic addressed in the interview.

## HOW IT WORKS (AUTOMATICALLY)

- **Data Analysis**
  - Collection of interview transcripts
  - Behavioral Pattern Extraction

- **Clustering Methodology**
  - Using K-means algorithms for cluster identification

- Identifying key points from conversations
- Analysis of responses for each type of interview

- Automatically determine the optimal number of clusters
- Analysis of similarity between responses
- Evaluation of internal cohesion of clusters

## OUTPUT

➔ **For each segment, the following are provided:**

- Detailed description of the target
- Behavioral Insights
- Points of anxiety and worries
- Competitive solutions used
- Specific recommendations

➔ **Comparative analysis**

- Common trends across clusters
- Significant differences
- Behavioral Patterns
- Key Decision Factors

## USE

### This bottom-up approach allows for

- Identify market segments naturally
- Understanding the specific needs of each group
- Developing personalized strategies
- Optimize resource allocation
- Improve the effectiveness of the proposed solutions

The resulting segmentation is then based on real data and observed behaviors, rather than a priori assumptions, ensuring greater accuracy and relevance for strategic decisions.

04

# Experiences and examples

**OF TESTS**

From Intent to Setup to Result

# 01

INTENT

Understanding the impact and value of smart features in a robot lawnmower

SETUP

| TYPE OF TEST | PARTICIPANTS | INPUT TEXT |
|---|---|---|
| Custom | Potential buyers of robot lawnmowers | How much more would you be willing to pay for a robot lawnmower with smart features like yard mapping, app control, and voice assistant integration? Which of these smart features do you consider most important and why? Are there other smart features you would expect to find in a high-end robot lawnmower? |

**1. Intent:** To understand the impact and value of smart features in a robotic lawnmower.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): Potential buyers of robot lawnmowers
  - Test input text (transforming intent into appropriate input for Aipermind): "How much more would you be willing to pay for a robot lawnmower with smart features such as yard mapping, app control, and voice assistant integration? Which of these smart features do you consider most important and why? Are there other smart features you would expect to find in a high-end robot lawnmower?"

**2. Intent:** To test prototypes of advanced safety systems for robotic lawnmowers.

- **Setup**
  - Test type: Solution validation
  - Participants (Digital Twin to be involved in the test): Parents with small children and/or pet owners
  - Test input text (transforming intent into appropriate input for Aipermind): "What are your main concerns about the safety of a robotic lawnmower, considering the presence of children or pets in your yard? What do you think about an advanced sensor system that detects the presence of obstacles and automatically stops the robot? Would you be willing to pay more for a robotic lawnmower with these additional safety features?"

**3. Intent:** To deepen the user experience with different control interfaces for the robot lawn mower.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): People with different levels of familiarity with technology
  - Test input text (transforming intent into appropriate input for Aipermind): "Would you prefer to control the

robot lawnmower via a built-in display on the device or via a mobile app on your smartphone? What advantages and disadvantages do you see in each option? How important is the ease of use and intuitiveness of the control interface to you?"

**4. Intent:** To evaluate the importance of design and aesthetics in a robotic lawnmower.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): People with a strong interest in design and outdoor furniture
  - Test input text (transforming intent into appropriate input for Aipermind): "How important is the aesthetics of a robotic lawnmower to you? Would you prefer a modern, minimalist design or a more traditional design? Which colors and materials do you think are best suited for a robotic lawnmower?"

**5. Intent:** Determine whether consumers are willing to accept aesthetic compromises in exchange for greater sustainability.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): Consumers with a strong interest in sustainability and ecology
  - Test input text (transforming intent into appropriate input for Aipermind): "Would you be willing to accept a robot lawnmower with a less "perfect" appearance (e.g. recycled plastic with small imperfections) if that meant it was more environmentally friendly and sustainable? How important is sustainability to you when choosing a robot lawnmower?"

**6. Intent:** To investigate user preferences regarding the acoustic feedback of the robotic lawnmower.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): People sensitive to noise or with particularly demanding neighbors
  - Test input text (transforming intent into appropriate input for Aipermind): "How important is the quietness of a robot lawnmower to you? Would you prefer a completely silent robot lawnmower or are you willing to accept a certain level of noise? What types of sounds made by the robot lawnmower bother you the most?"

**7. Intent:** To explore the acceptance of different configurations for robot lawnmower charging stations.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): People with gardens of different sizes and shapes
  - Test input text (transforming the intent into a suitable input for Aipermind): "Would you prefer a discreet and inconspicuous charging station for the robotic lawnmower or a charging station that integrates into the garden design? How important is the ease of installation and convenience of the charging station to you?"

**8. Intent:** To test the acceptance of a robotic lawnmower with a minimalist design.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): Minimalist design and technology enthusiasts
  - Test input text (transforming intent into appropriate input for Aipermind): "What do you think of a lawnmower robot with an extremely simple and minimalist design, without any decorative or flashy elements? Would you find it attractive or would you prefer a more elaborate design?"

**9. Intent:** To gather feedback on different types of materials and finishes for the body of the robot lawnmower.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): People interested in product durability and maintenance
  - Test input text (transforming intent into appropriate input for Aipermind): "Would you prefer a

scratch-resistant plastic case or a more elegant metal case? How important is the ease of cleaning and weather resistance of the case material to you?"

**10. Intent:** To investigate user preferences regarding the color range available for the robotic lawnmower and accessories.

- **Setup**
  - Test type: Custom
  - Participants (Digital Twin to be involved in the test): People who want to customize the look of their robot lawnmower
  - Test input text (transforming intent into appropriate input for Aipermind): "What colors would you prefer for your robot lawnmower? Would you be interested in colorful or customizable accessories for the robot lawnmower? How important is it for you to be able to express your personal style through the robot lawnmower?"

**11. Intent:** To compare user perceptions of robotic lawnmowers made of different materials.

- **Setup**
  - Test type: Comparison
  - Participants: Consumers interested in purchasing a robot lawnmower
  - Test input text: "We will show you two robot lawnmowers, one made of recycled plastic and the other of aluminum. Which of the two do you think is of superior quality? Which of the two would you consider more sustainable? Which of the two would you buy and why?"

**12. Intent:** To evaluate the importance of the finishes on the body of the robot lawnmower.

- **Setup**
  - Test type: Comparison
  - Participants: Consumers interested in purchasing a robot lawnmower
  - Test input text: "We'll show you two robot lawnmowers, one with a matte finish and the other with a glossy finish. Which one do you prefer aesthetically? Which one do you think is easier to clean? Which one would you buy and why?"

**13. Intent:** To test the visual impact and perceived quality of a robotic lawnmower with a seamless design.

- **Setup**
  - Test type: Comparison
  - Participants: Consumers interested in purchasing a robot lawnmower
  - Test input text: "We will show you two robot lawnmowers, one with visible seams in the body and the other without seams. Which one seems more modern to you? Which one seems higher quality to you? Which one would you buy and why?"

**14. Intent:** To investigate the importance of the manufacturer's logo on the robot lawnmower.

- **Setup**
  - Test type: Custom
  - Participants: Consumers interested in purchasing a robot lawnmower
  - Test input text: "How important is it to you that the robotic lawnmower has the manufacturer's logo clearly visible? Would you prefer a large, eye-catching logo or a small, discreet logo? Where would you prefer the logo to be placed?"

**15. Intent:** To compare different design solutions for the robot lawnmower.

- **Setup**
  - Test type: Comparison
  - Participants: Consumers interested in purchasing a robot lawnmower
  - Test input text: "We will show you two robot lawnmowers, one with rounded shapes and the other with angular shapes. Which of the two do you prefer aesthetically? Which of the two seems more modern to you? Which of the two would you buy and why?"